

STA 2270 STATISTICS

Lecture Notes

Dr. Mutua Kilai, PhD

Table of contents

Introduction	3
Types of Variables	3
Sources of Data	3
Data Presentation	4
Frequency Distribution Tables	4
Construction of Ungrouped Frequency Distribution	4
Construction of Grouped Frequency Distribution	4
Graphical Displays	6
Pie Chart	7
Bar Chart	8
Pareto Charts	8
Histogram	10
Frequency Polygon	10
Cumulative Frequency Curve	12
Numerical Summaries	14
Measures of Location	14
Measures of Central Tendency	14
Measures of Spread/Dispersion	21
Measures of Relative Dispersion	28
Measures of Shape	30
Skewness	30
Kurtosis	34
Bivariate Data	36
Introduction	36
Scatter Diagrams and Correlation	36
Positive and Negative Correlation	36
Interpreting a Scatter Plot	37
Correlation Coefficient	37
Pearson's Product Moment Correlation Coefficient	37
Spearman's rank correlation coefficient	39
Regression	40
Regression Equation	40
Simple Linear Regression	40
Least Squares Estimation of the Parameters	41
Estimation of β_0 and β_1	41
Probability	44
Definitions	44
Rules of Probability	44
Counting Rules Useful in Probability	47
Permutations	47
Combinations	47
Conditional Probability	48

Discrete Probability Distributions	49
Introduction	49
Some Special Distributions	54
Binomial Distribution	54
Poisson Distribution	55
Basic Concepts of Hypothesis Testing	56
One-Tailed vs. Two-Tailed Tests	57
Example 1: Two-Sample Independent t-Test	58
Example 2: One-Tailed Z-Test for a Population Proportion	59

Introduction

i Definition 1: Statistics

A branch of science that deals with collection presentation, analysis, and interpretation of data.

The definition points out 4 key aspects of statistics namely:

- i. Data Collection
- ii. Data Presentation
- iii. Data Analysis
- iv. Data Interpretation

Statistics is divided into two broad categories namely **descriptive** and **inferential** statistics

Descriptive Statistics: Summary values and presentations which gives some information about the data.

Inferential Statistics: summary values calculated from the sample in order to make conclusions about the target population.

Types of Variables

There are 2 broad categories namely qualitative and quantitative variables.

Qualitative Variables Variables whose values fall into groups or categories. They are called categorical variables and are further divided into 2 classes namely nominal and ordinal variables

1. **Nominal Variables:** variables whose categories are just names with no natural ordering. Eg gender marital status, skin colour, district of birth etc
2. **Ordinal Variables:** variables whose categories have a natural ordering. Eg education level, performance category, degree classifications etc

Quantitative Variable these are numeric variables and are further divided into 2 classes namely discrete and continuous variables

1. **Discrete variables:** can only assume certain values and there are gaps between them.
2. **Continuous variables:** can assume any value in a specified range.

Sources of Data

There are 2 sources for data namely Primary, and Secondary data:

Primary data: - freshly collected. They are original in character ie they are the first hand information collected, compiled and published for some purpose. They haven't undergone any statistical treatment.

Secondary Data:- 2nd hand information mainly obtained from published sources such as statistical abstracts books encyclopaedias periodicals, media reports eg census report CD-roms and other electronic devices, internet. They are not original in character and have undergone some statistical treatment at least once.

Data Presentation

Frequency Distribution Tables

Definition (Raw Data): unprocessed data ie data in its original form.

Frequency Distribution: The organization of raw data in table form with classes and frequencies. Rather it's a list of values and the number of times they appear in the data set. We have grouped and ungrouped frequency distribution tables for large and small data sets respectively.

Construction of Ungrouped Frequency Distribution

- Note the largest and smallest observations in the data.
- Starting with the smallest value, tally the observations of each quantity.
- Count the number of tallies for each quantity and record it as frequency.

Example

Construct an ungrouped frequency table for the data below 16 14 15 13 12 14 16 15 15 14 17 16 13 16 15 14 18 13 15 17

Solution

Value	12	13	14	15	16	17	18
Frequency	1	3	4	5	4	2	1

Construction of Grouped Frequency Distribution

When the number of observations is too large and/or when the variable of interest is continuous, it's cumbersome to consider the repetition of each observation.

A quick and more convenient way is to group the range of values into a number of exclusive groups or classes and count the class frequency.

The resulting table is called a grouped frequency distribution table.

A grouped frequency distribution consists of classes and their corresponding frequencies.

Each raw data value is placed into a quantitative or qualitative category called a class.

Steps in construction

1. Select the number of classes k . Choose the smallest integer such that

$$2^k > n \Rightarrow k > \frac{\log n}{\log 2}$$

2. Identify the largest and smallest observation and compute the range (R).
3. Identify the smallest unit of measurement (u) used in the data collection (ie the accuracy of the measurement.)

For data measurement: 10, 30, 20, 50, 30, 60, $u = 10$

For data measurement: 12, 15, 11, 17, 13, $u = 1$

For data measurement: 1.6, 3.2, 2.8, 5.6, 3.5, 1.6, $u = 0.1$

4. Estimate the class width/interval i round up $\frac{R}{k}$ to the nearest u . E.g round up 3.13 to the nearest 0.1 is 3.2
5. Pick the starting value (lower class limit of the 1st class (LCL1)) as the smallest value used in the computation of R above. Successive LCLs are got by adding i to the previous LCL
6. Find the upper class limit of the 1st class (UCL1)) by subtracting u from LCL2. Successive UCLs are got by adding i to the previous UCL
7. If necessary find the class boundaries as follows:

$$LCB = LCL - \frac{1}{2}u, UCB = LCL + \frac{1}{2}u$$

8. Tally the number of observations falling in each class and record the frequency. A value x fall into class LCL-UCL if $LCB \leq x \leq UCB$

Example

Organize the data below into a grouped frequency table. 15.0 17.4 10.3 9.2 20.7 18.9 16.6 22.4 23.7 18.6 26.1 16.5 19.7 12.9 15.7 30.8 15.4 20.3 24.0 29.6 18.3 23.7 17.8 24.6 23.0 21.4 32.8 12.5 17.5 18.3 23.2 21.6 20.8 29.8 24.5 28.4 13.5 17.1 27.1 27.9

Solution

$$u = 0.1, n = 40, \Rightarrow k > \frac{\log 40}{\log 2} \Rightarrow k = 6$$

$$\text{The range: } 32.8 - 9.2 = 23.6$$

$$i = \text{round up } \frac{23.6}{6} \text{ to the nearest } 0.1 = 4.0$$

$$\text{Now } LCL_1 = 9.2 \Rightarrow LCL_2 = 13.2, LCL_3 = 17.2$$

$$UCL_1 = LCL_2 - u = 13.2 - 0.1 = 13.1$$

The frequency table is shown below:

Class	Boundaries	Frequency	CF
9.2-13.1	9.15 - 13.15	4	4
13.2 - 17.1	13.25 - 17.15	7	11
17.2 - 21.1	17.25 - 21.15	11	22
21.2 - 25.1	21.25 - 25.15	10	32
25.2 - 29.1	25.25 - 29.15	4	36
29.2 - 33.1	29.25 - 33.15	4	40

Exercise

1. These data represent the record high temperatures in degrees Fahrenheit (F) for each of the 50 states. Construct a grouped frequency distribution for the data. 112 100 127 120 134 118 105 110 109 112 110 118 117 116 118 122 114 114 105 109 107 112 114 115 118 117 118 122 106 110 116 108 110 121 113 120 119 111 104 111 120 113 120 117 105 110 118 112 114 114
2. Suppose a researcher wished to do a study on the ages of the top 50 wealthiest people in the world. The researcher first would have to get the data on the ages of the people. In this case, these ages are listed in Forbes Magazine. 49 57 38 73 81 74 59 76 65 69 54 56 69 68 78 65 85 49 69 61 48 81 68 37 43 78 82 43 64 67 52 56 81 77 79 85 40 85 59 80 60 71 57 61 69 61 83 90 87 74 Organize the data into a grouped frequency table

Graphical Displays

After you have organized the data into a frequency distribution, you can present them in graphical form.

The purpose of graphs in statistics is to convey the data to the viewers in pictorial form.

It is easier for most people to comprehend the meaning of data presented graphically than data presented numerically in tables or frequency distributions.

This is especially true if the users have little or no statistical knowledge.

Statistical graphs can be used to describe the data set or to analyze it.

Graphs are also useful in getting the audience's attention in a publication or a speaking presentation. They can be used to discuss an issue, reinforce a critical point, or summarize a data set.

They can also be used to discover a trend or pattern in a situation over a period of time.

The commonly used graphs in research are; the pie chart, bar chart, histogram, frequency polygon and the cumulative frequency curve (ogive).

! Important

We will demonstrate this from a practical perspective using R. Ensure you familiarize yourself with R Programming language.